

Richard Lin

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EDUCATION

Temple University Philadelphia, PA
Bachelor of Science in Electrical and Computer Engineering *Diploma Expected: May 2028*
Minor: Computer Science | Fundamentals of Physics Certificate | GPA: 3.95/4.00 | Dean's List: F24, S25, F25
Relevant Courses: Differential Equations; Digital Circuit Design; Circuits & Electronics I-II; Computation I-II

PROFESSIONAL EXPERIENCE

Temple University Philadelphia, PA
Undergraduate Research Assistant – Computer Fusion Lab (CFL) *Oct 2025 – Present*
Advisor: Dr. Li Bai

- Administered lab website and systems; ensured security, software maintenance and network reliability
- Developed an AI assistive library with recommendation system using Python, LLMs, multi-agent workflows, and RAG to classify 400+ devices into 66 categories and natural-language accessibility matching

Undergraduate Teaching Assistant *Aug 2025 – Present*

- Assisted faculty with instruction, activities, assignment evaluation, and preparation of course materials
- ENGR 1101:** Mentored students in embedded system, programming, circuits with Micro:bit
- ENGR 1102:** Supported engineering design and data analysis using Excel, Python, NumPy, and Matplotlib
- ECE 1111:** Taught programming fundamentals including C, C++, Python, and Linux/Bash

PROJECTS

Musical Light Show – ECE 2352: Circuits and Electronics II *May 2026*

- Designed analog audio-processing system that visualizes music audio-reactive LED light show
- Built op-amp amplifier, adjustable gain control, high and low-pass filters, and BJT LED driver circuits
- Validated design and performance through Multisim simulation and hardware testing

Detective Conan AI Glasses (Childhood Dream Project) *June 2026*

- Designed and built AI smart glasses using Raspberry Pi Zero 2 W and ESP32 with camera, microphone, solar-assisted charging, and a custom 3D-printed enclosure
- Developed voice assistant integrating speech recognition, local/cloud LLMs, and text-to-speech
- Implemented object detection, OCR, face recognition, translation, navigation, and cybersecurity monitoring

CLUBS & LEADERSHIP

Sustainable Temple Energy and Power Scholars (STEPS) Philadelphia, PA
STEPS Ambassador *Sep 2024 – Present*

- Led STEM demos promoting engineering and energy for HS students; mentored STEPS underclassmen

Institute of Electrical and Electronics Engineers (IEEE) – Temple University Philadelphia, PA
President *Sep 2024 – Present*

- Led a team of ~11 officers in branch initiatives; outreach, workshops, and student engagement activities
- Increased IEEE SAC conference participation by 70% through outreach and IEEE membership engagement
- Led a multi-university hardware hackathon with participation from Drexel and UPenn IEEE

Temple University Eta Kappa Nu (IEEE-HKN) Philadelphia, PA
Secretary *June 2026 – Present*

- Coordinate member communications and maintain records for professional development, outreach, and chapter events.

Temple University College of Engineering Alumni Association Philadelphia, PA
Student Representative *June 2026 – Present*

SKILLS

Programming: Python, C, C++, Java, JavaScript, Bash, HTML/CSS, Verilog/SystemVerilog, MATLAB
AI: LLM integration (Ollama, Cloud API), Prompt Engineering, Computer Vision (OpenCV, YOLO), Text-to-Speech, OCR
Tools: Git/GitHub, Linux, Raspberry Pi, n8n, FPGA (DE-Lite 10), MultiSim, Arduino, PyTorch, MS Office, AutoCAD
Skills: Hardware-Software integration, Multi-Agent Systems, UI/UX development, Circuit Design & Analysis, Soldering
Languages: English, Mandarin Chinese, Fuzhounese, Spanish